Door Stops at NIT-Goa: the cause and its solution!

A **doorstop** is any device, wedge, or heavy object used to keep a door open, hold it closed, or prevent it from swinging too far and damaging a wall or furniture.



Figure 1: image of a door stop in general

1 Field Investigation of door stops in campus

I took some images of the door stops at Gyan Mandir.





Figure 2 and 3: door stops at NIT-Goa campus

Common classrooms doors in NIT Goa are about 81 inches by length. Its width is around 35 inches. The doors are 1.5 inch thick.

As per research, the mass of a common door in this college is around 25kg.



Figure 4: common door image

As per research, the dimensions of the door stop are as follows:

No.	Component	Dimension	Unit
1	Overall length	80	mm
2	Mounting plate length	55–65	mm
3	Mounting plate width	35–45	mm
4	Rod diameter	8–10	mm
5	Distance between rods	20–25	mm
6	Rubber head width	40–45	mm
7	Rubber head length	30–35	mm

Figure 5: table of dimensions of door stop

By measurement, the center of the plate of the door stop, is usually 2 inches away from the base of the door, and 3.2 inches away from the vertical end of the door.
It should be noted that the length of the protruding arm of the stop is 80 to 100 mm max.

The door stop arm is made such that it can rotate freely 180 degree.

The plate and rod is made of Stainless Steel (SS 304), while the rubber end is made of Synthetic Rubber (EPDM or NBR).
The rubber material is such that it works efficiently on marble tiles. It is intended that when the user foots the arm down, the rubber must rub against the tile and cause the door to stop.

Parameter	Value
Frequency of operation	20–30 cycles/day
Average daily operations	25 cycles/day
Annual operations	9,125 cycles/year
Design life assumption	10 years
Total lifetime operations	91,250 cycles

Figure 6: operation data for span of 10 years.

It is observed that the arm rotates best on summer and winters, but many of the arms get tight during monsoon, making the steel irrelevant.
Almost every door stopper at NIT Goa is tight and does not rotate, and the main reason could be a issue with the shape and product itself.
It should also be observed that the door stop is filthy dirty with dust.

No.	Component	Failure mode	Possible cause
1	Rubber head	Rubber wear or cracking	Repeated impact and ageing
2	Metal rods	Bending or deformation	Excessive impact force from the door
3	Mounting plate	Loosening	Repeated vibrations and improper fastening
4	Screws	Loosening or corrosion	Repeated loading and moisture exposure
5	Metal body	Corrosion	Exposure to moisture and humidity
6	Rubber head	Detachment	Weak bonding or repeated impact

Figure 7: Observed / Potential Failure Modes of Double-Rod Door Stopper

2 Load Characterization

Actual Loading Conditions

1. Normal operation: The door gradually contacts the rubber head, producing compressive and bending loads.
2. Sudden door opening: A rapidly moving door produces a higher impact force.
3. Repeated operation: Continuous opening and closing results in cyclic loading and possible fatigue.
4. Overloading: Excessive force applied to the door may cause permanent bending of the rods or loosening of the mounting screws.
5. Environmental exposure: Moisture and humidity may cause corrosion of the metallic components and deterioration of the rubber.

Critical loading condition

The most critical loading condition is the sudden impact load produced when the moving door strikes the rubber head. This generates compression in the rubber, bending in the supporting rods, and forces on the mounting plate and screws.

No.	Type of load	Description	Component affected
1	Impact load	Sudden force when the moving door strikes the rubber stopper.	Rubber head and metal rods
2	Compressive load	Compression of the rubber head due to contact with the door.	Rubber head
3	Bending load	Bending moment generated on the rods due to the force applied by the door.	Metal rods
4	Shear load	Shear forces developed at the rod connections and mounting plate.	Rods and mounting plate
5	Tensile load	Pulling forces acting on the mounting screws due to the overturning moment.	Mounting screws
6	Fatigue loading	Repeated loading and unloading during daily door operation.	Metal rods, mounting plate and screws
7	Frictional load	Friction between the door surface and the rubber head during contact.	Rubber head

Figure 8: Significant Loads and Loading Conditions Experienced by the Door Stopper

Calculations:

Parameter	Symbol	Measured value
Door mass	m	25 kg
Door width	L	889 mm
Hinge-to-stopper distance	r	800 mm
Angular velocity	ω	Not measured
Rod diameter	d	9 mm
Rod unsupported length	l	80 mm

Figure 9: measurements considered

A. Moment of inertia of the door

Assuming the door is a uniform rectangular plate rotating about its vertical hinge axis:

$$I = \frac{1}{3}mL^2$$

$$I = \frac{1}{3}(25)(0.889)^2$$

$$I = 6.586\ kgm^2$$

B. Distance ratio

$$\frac{r}{L} = \frac{0.800}{0.889} = 0.9$$

C. Rod cross-sectional area

Assuming both rods have a circular cross-section:

$$A = \frac{\pi d^2}{4}$$

$$A = \frac{\pi(9)^2}{4} = 63.62\ mm^2$$

For two identical rods: $A_{total} = 127.23\ mm^2$

D. Rod bending characteristics

For a circular rod, the second moment of area is:

$$I_r = \frac{\pi d^4}{64} = 322.06\ mm^4$$

The section modulus is:

$$Z = \frac{\pi d^3}{32} = 71.56\ mm^3$$

3 Failure Analysis

Component	Potential failure mechanism	Engineering explanation
Rubber head	Impact damage	Repeated collision with the door can cause cracking, tearing or permanent deformation.
Rubber head	Excessive compressive stress	High contact pressure can cause permanent compression or damage.
Metal rods	Bending failure	Impact force acting away from the rod support creates a bending moment.
Metal rods	Fatigue failure	Repeated bending cycles can initiate and propagate fatigue cracks.
Metal rods	Excessive stress	High bending or axial stress may exceed the material's allowable strength.
Rod connections	Shear failure	Transverse forces can produce shear stress at the rod-to-plate connections.
Mounting plate	Bearing failure	Contact pressure around screw holes can cause local deformation or hole elongation.
Mounting screws	Tensile failure	An overturning moment can generate tensile forces in the mounting screws.
Mounting screws	Loosening due to cyclic loading	Repeated impact and vibration can reduce fastening security.
Metal body	Stress concentration	Changes in cross-section, sharp corners and holes can produce localized high stresses.
Metal body	Corrosion	Moisture exposure can reduce the effective cross-sectional area over time.

Figure 10: list of failures

When the moving door strikes the rubber head, the force is transferred to the supporting rods. Since the rods extend from the mounting plate, the impact can generate a bending moment.

For a simplified cantilever model:

$$M = Fl$$

The maximum bending stress is:

$$\sigma_b = \frac{32Fl}{\pi d^3}$$

A. Bending failure of metal rods

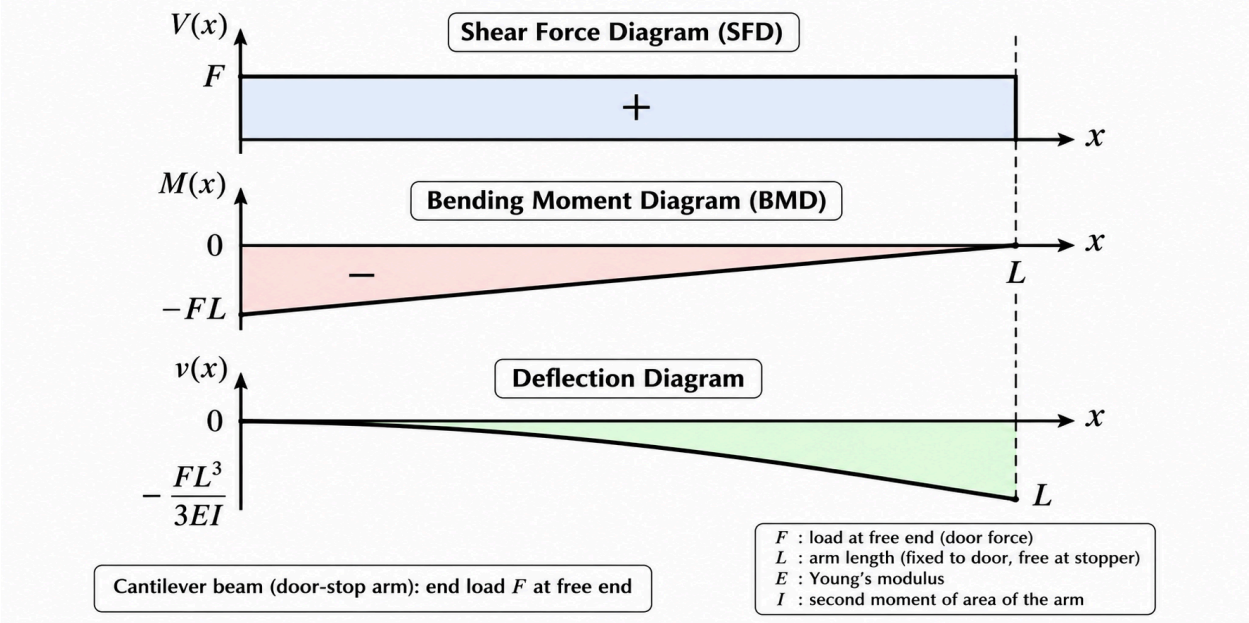


Figure 11: SFD and BMD diagram

B. Shear failure

For two rods sharing the load equally, the average direct shear stress would be:

$$\tau = \frac{F}{2A}$$

C. Bearing failure

Bearing stress may develop where the mounting screws contact the holes in the mounting plate.

The projected bearing stress can be estimated using:

$$\sigma_{br} = \frac{F}{td}$$

where:

- F = force transferred through the screw
- t = mounting plate thickness
- d = screw diameter

D. Impact failure

Impact is the primary loading event during normal operation.

When the moving door contacts the rubber head:

1. The rubber undergoes compression.
2. The impact force is transferred to the supporting rods.
3. The rods experience bending and potentially shear.
4. The mounting plate transfers the resulting loads to the floor.

E. Fatigue failure

Fatigue occurs when a component experiences repeated loading and unloading over an extended period.

The rods and their connections are subjected to repeated bending and impact loading during door operation.

Potential consequences include:

- Initiation of small cracks.
- Crack propagation.
- Progressive weakening of the metal.
- Eventual fracture.

4 Solution Model

I came across this image, which striked an idea into me. The figure below is the image that gave me the idea:



figure 12: initial idea

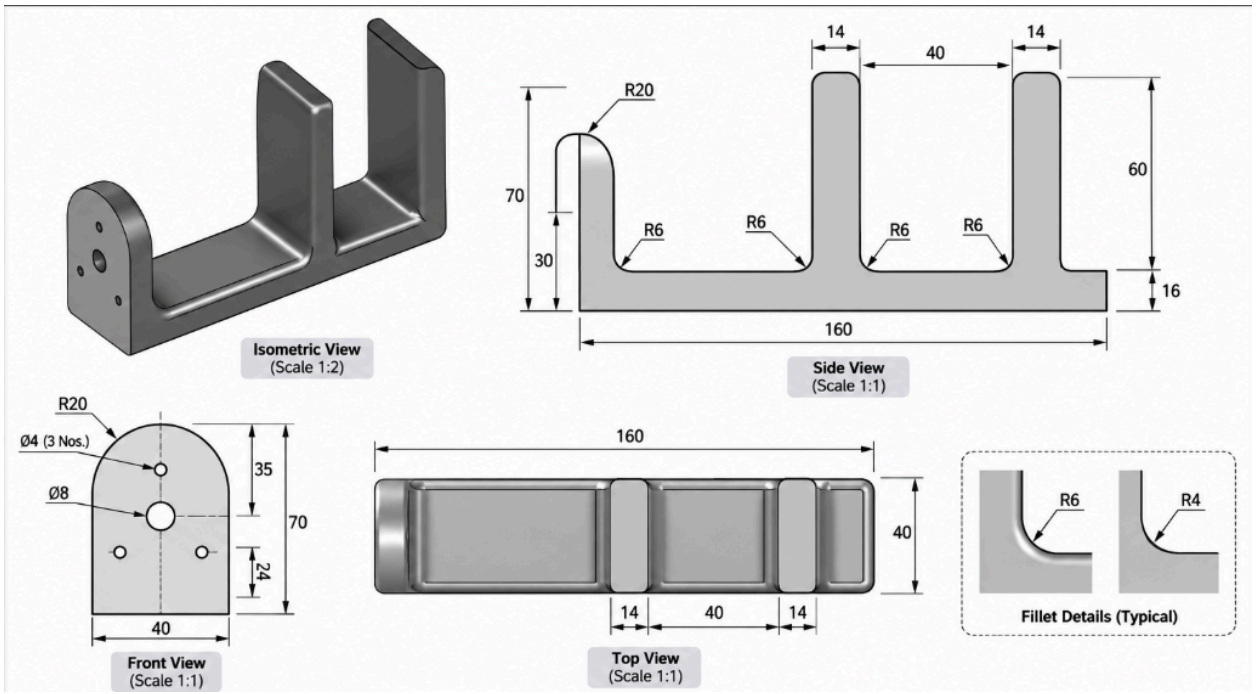


Figure 13: My final idea which i made using the door parameters in our college.

The image above is the top, side and top view along with the 3D image which show the proposed dimensions of my project.
Following are the snaps of the CAD file i made using fusion 360.

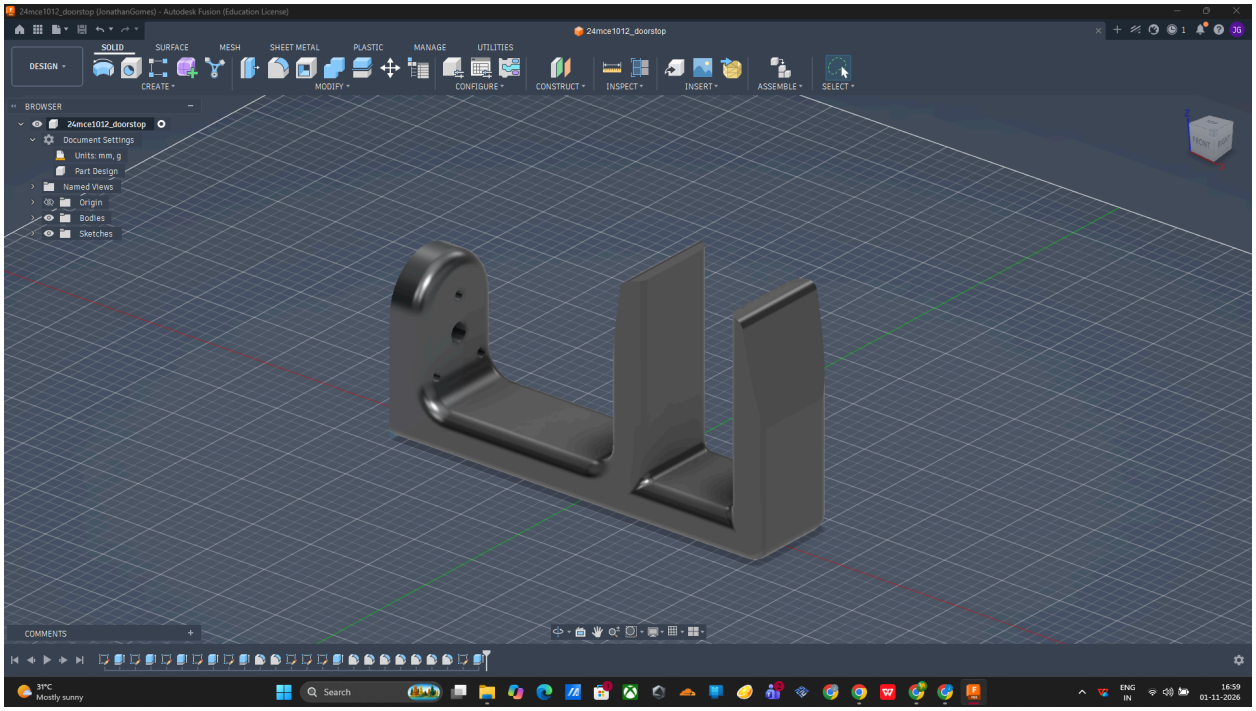


Figure 14: view 1

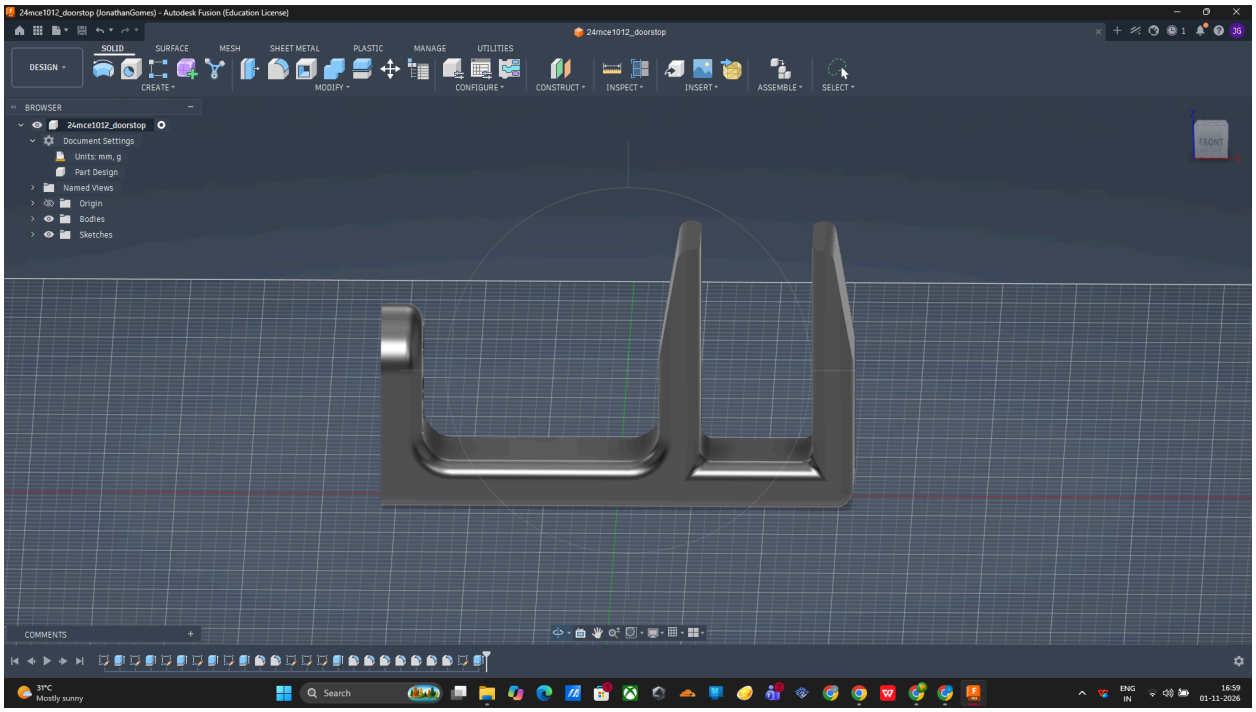


Figure 15: view 2

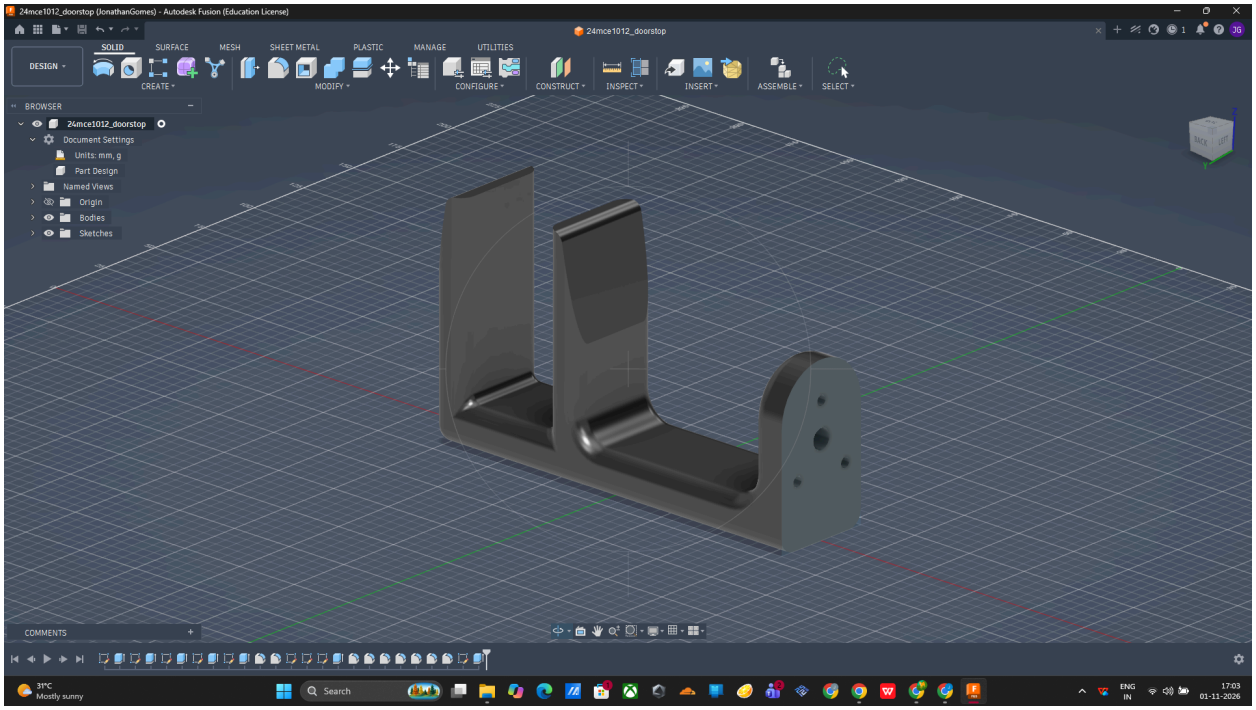


Figure 16: View 3

Working:

I gave the arms a slope to make it easier to fit it in and pull out. The length of arms from the screwed base is given such that the tpu model can be stretched for easy use. The gap between arms is also given so it can accommodate the door without much shaking. The function of all this is so that the door remains firm and does not get pulled out unless someone stretches arms out.

Engineering calculations:

1 Door moment of inertia

We first determine the rotational inertia of the existing door.

Assuming a uniform rectangular door rotating about its vertical hinge axis:

$$I_d = \frac{1}{3} m L^2$$

Substituting:

$$I_d = \frac{1}{3}(25)(0.889)^2 = 6.586\ kgm^2$$

4. Rotational kinetic energy before impact

The kinetic energy of the rotating door immediately before contacting the stopper is:

$$KE = \frac{1}{2} I_d \omega^2 = 3.293\omega^2\ \text{J}$$

5. Mechanical analysis of the two supporting arms

A. Second moment of area

For a rectangular cross-section:

$$I_a = \frac{bt^3}{12}$$

$$I_a = \frac{40(14)^3}{12} = 9146.67\ mm^3$$

B. Section modulus

$$Z_a = \frac{bt^2}{6}$$

$$Z_a = \frac{40(14)^2}{6} = 1306.67\ mm^3$$

C. Bending moment

For a horizontal force P acting at the top of an arm:

$$M = P \times h$$

$$M = P \times 75 = 75P$$

D. Maximum bending stress

$$\sigma_b = \frac{M}{Z_a}$$

$$\sigma_b = \frac{75P}{1306.67} = 0.05740 \text{ MPa}$$

If both arms share the total applied force equally:

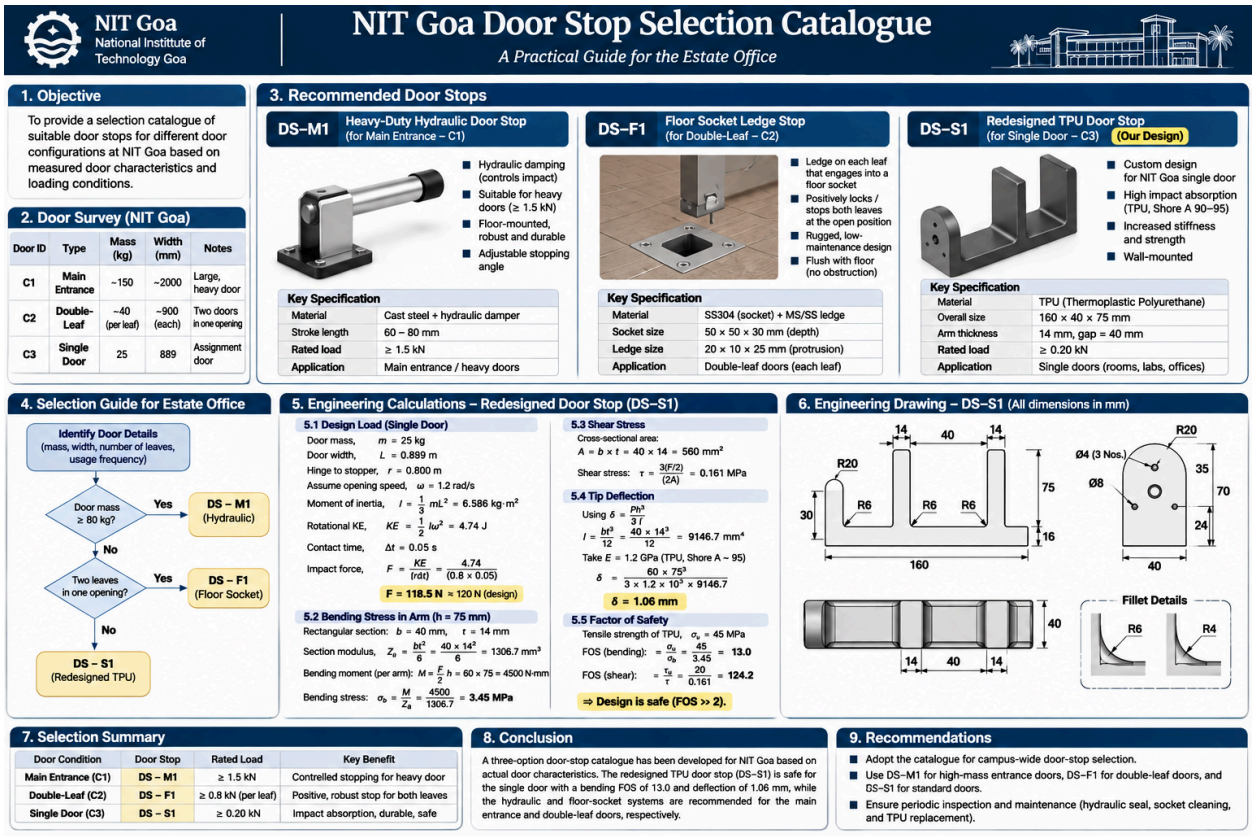
$$P = \frac{F}{2} = 0.02870 \text{ MPa}$$

Why my project is theoretically better?

Parameter	Original	Redesigned
Second moment of area	322.06 mm^4	9146.67 mm^4
Section modulus	71.56 mm^3	1306.67 mm^3
Improvement in I	—	28.4 times
Improvement in Z	—	18.3 times

Higher moment of inertia and section modulus means the door stop will be better at handling higher impact.thus my project can handle the impact of wind better than the normal door stop.

5 Selection Catalogue



1. Purpose and scope

This catalogue is a selection framework for three door configurations identified by the student during the NIT Goa field survey. It combines measured data available for the single-door redesign with a survey checklist for the other two door configurations. Where field measurements or manufacturer ratings are unavailable, they are explicitly marked as pending rather than invented.

Code	Door configuration	Catalogue selection
C1 / DS-M1	Giant main entrance	Commercial heavy-duty door-control/stop assembly with damping; final model to be selec
C2 / DS-F1	Two leaves in one opening	Retractable door-mounted pin/ledge engaging a recessed floor socket; one independent s
C3 / DS-S1	Single door used for redesign	Custom wall-mounted two-arm TPU stop based on the supplied CAD dimensions.

2. Field survey data

Parameter	C1 — Main entrance	C2 — Twin-leaf	C3 — Single door
Known configuration	Giant main door	Two doors/leaves in one opening	Single door
Mass	Not measured	Not measured (each leaf)	25 kg
Width	Not measured	Not measured (each leaf)	889 mm
Hinge-to-stop distance	Not measured	Not measured	800 mm
Opening speed / cycles	Not measured	Not measured	Not measured

Parameter	C1 — Main entrance	C2 — Twin-leaf	C3 — Single door
Mounting substrate	To inspect	Floor slab and door stile to inspect	Wall substrate / anchors to verify

The earlier catalogue concept contained illustrative values for C1 and C2. Those values are not field measurements and are not used as engineering design inputs in this revised edition.

3. Catalogue product entries

DS-M1 — Heavy-duty damped stop for the giant main entrance (C1)

Recommended product class: a commercially rated heavy-duty door-control or stopping assembly with hydraulic/mechanical damping, selected to suit the actual door closer, opening angle, mounting substrate and usage. Before approval, obtain the supplier's allowable door mass/width, stopping energy or force rating, cycle rating, adjustment range, installation drawing and anchor requirements. No numeric capacity is assigned in this catalogue without those documents.

Selection item	Requirement
Material / construction	Supplier-specified corrosion-resistant metal body and damping element.
Mounting	Floor, wall or overhead only as permitted by product instructions and site conditions.
Verification	Compare documented product rating and anchor capacity with measured door demand.
Use case	Large entrance doors where controlled stopping and impact damping are required.

DS-F1 — Recessed floor-socket ledge stop for the twin-leaf doorway (C2)

Each door leaf carries a retractable vertical pin or drop-bolt/ledge near its lower edge. At the desired open position, the pin is lowered into a stainless-steel receiver socket recessed flush into the floor. The socket provides positive mechanical engagement and prevents further swing. Use one independent pin-and-socket set for each leaf unless the surveyed arrangement justifies a different engineered solution.

Part	Design intent / requirements
Door-mounted pin/ledge	Guided retractable corrosion-resistant pin; secure housing fixed to the door stile; pin must fully retract before the leaf mo
Floor receiver	Recessed stainless-steel socket/liner with rounded lead-in, clean-out access and drainage provision where needed.
Position	Locate socket to achieve required open angle and adequate engagement; ensure the two leaves and handles do not coll
Floor integration	Receiver must be flush with finished floor; check slab reinforcement, embedded services, waterproofing and accessibility
Maintenance	Inspect pin return, socket debris/corrosion, fasteners and engagement depth.

Operating sequence: (1) open the leaf to its designated position; (2) align pin over receiver; (3) lower pin into socket; (4) retract pin fully to release before closing.

DS-S1 — Custom TPU door stop for the surveyed single door (C3)

The student's custom design is a wall-mounted, two-arm TPU stopper. The design geometry below uses the supplied dimensions. Its final allowable load and safety factor remain subject to the missing impact measurements, exact TPU grade, manufacturing process and mounting verification.

5.3 Arm cross-section properties

For the preliminary rectangular section, b = 40 mm and t = 14 mm, with 14 mm in the bending direction:

Area A = bt = 40 × 14 = **560 mm²**.
Second moment I_a = bt³/12 = 40 × 14³/12 = **9,146.67 mm■**.
Section modulus Z_a = bt²/6 = 40 × 14²/6 = **1,306.67 mm³**.

The actual CAD orientation must match this assumed bending axis. If the section is oriented differently, I and Z must be recalculated.

5.4 Bending stress for 75 mm arm height

Model one arm as a cantilever with horizontal tip force P. M = Ph = 75P N·mm. Therefore σ_b = M/Z = 75P/1306.67 = **0.05740P MPa**, with P in newtons. If two arms share total force F equally, P = F/2 and σ_b = 0.02870F MPa. Equal load sharing must be demonstrated by the contact geometry; also check the conservative case where one arm takes the full load.

5.5 Transverse shear stress

For a rectangular section, τ_{max} = 1.5P/A = 1.5P/560 = **0.002679P MPa**. This is a nominal beam-theory value; the arm-to-base junction needs a local stress check.

5.6 Tip deflection

For a cantilever with an end load, δ = Ph³/(3EI). With h = 75 mm and I = 9,146.67 mm■: δ = **15.37P/E mm**, where E is the actual TPU modulus in MPa and P is the force on one arm.

5.7 Base, wall plate and mounting

The base gross volume is 160 × 40 × 16 = **102,400 mm³** before subtracting fillets, holes and other features. Mounting-plate net-section, screw bearing/tension and anchor pull-out checks require actual plate thickness, hole coordinates, fastener type/grade, embedment, wall substrate and edge distances.

5.8 Stress concentration, fatigue and material

R6 and R4 fillets help reduce abrupt geometry changes, but the stress-concentration factor depends on local geometry and load direction. Fatigue life requires service cycles and grade-specific TPU cyclic data. Obtain the exact TPU supplier's stress-strain curve, modulus, tensile/flexural strength, tear resistance, Shore hardness, temperature limits and (if printed) build orientation/infill effects.

5.9 Factor of safety

A final factor of safety cannot yet be calculated because the applied design load and grade-specific allowable properties are unknown. Use FOS = allowable strength / calculated stress only after establishing a justified design load and appropriate TPU allowable at the service temperature and strain. Include creep, fatigue, manufacturing variation and mounting failure in the assessment.

6. DS-F1 engineering checks for the twin-leaf floor socket

Failure mode	Calculation / evidence required
Pin shear	τ = F/(nπd²/4); determine force per leaf, pin diameter and number of shear planes.
Pin bending	M = F·e; σb = M/Z; measure exposed pin length and support condition.
Socket bearing	p = F/(d·t_engagement); measure pin diameter and engagement depth.
Door attachment	Check housing fasteners and door-stile material/thickness, edge distances and screw pull-out.
Floor receiver and anchors	Check receiver flange, anchors, concrete/slab strength, embedment, edge distances and possible breakout.
Impact and fatigue	Measure leaf angular speed, stopping response and cycle frequency; test repeated engagements.

No DS-F1 dimensions or load capacity are assigned until the two leaf masses, widths, opening speeds, pin geometry, socket engagement and floor construction are measured.

7. Estate Office selection workflow

Stage	Action	Decision
1 — Identify	Record door mass, width, height, leaf count and hinge-to-contact distance	Door configuration.
2 — Observe	Measure/estimate opening speed, frequency, wind exposure and door-closer setting	Operating conditions.
3 — Inspect	Identify wall/floor substrate, available fixing locations, slab services and ac	Feasible mounting method.
4 — Calculate	Determine rotational energy, stopping reaction, pin/arm stresses and anch	Required design action.
5 — Select	Compare required action with documented manufacturer ratings or verified	Acceptable design capacity.
6 — Validate	Install prototype/product; inspect stopping position, deflection, fasteners a	Confirmed or revised selection.

8. Summary of catalogue recommendations

Door code	Door type	Recommended option	Critical verification
C1	Giant main entrance	DS-M1 — rated heavy-duty damped door-control/stop assembly.	Stop assembly supplier rating and anchor capacity.
C2	Twin-leaf doorway	DS-F1 — one retractable pin/ledge per leaf engaging a recessed floor socket	Pin shear/bending, socket bearing, floor anchorage and tri
C3	Single door (25 kg)	DS-S1 — custom two-arm TPU wall stop.	Impact force, TPU properties, arm stress/deflection, wall a

9. Inspection and maintenance recommendations

DS-M1: inspect damping performance, seals/leaks, fasteners and mounting base. DS-F1: keep sockets clean, check pin retraction and engagement, inspect corrosion and fasteners, and ensure the socket remains flush with the floor. DS-S1: inspect TPU for tearing, permanent deformation, cracks and loosening; replace if damage or excessive permanent set is observed.

10. Conclusion

This revised catalogue covers the three door configurations identified for NIT Goa. It recommends a rated damped door-control/stop system for the giant main entrance (DS-M1), a retractable pin engaging a recessed floor socket for each leaf of the twin-leaf doorway (DS-F1), and the student's custom TPU design for the measured single door (DS-S1). The single-door inertia and geometric section properties are calculated from supplied measurements. The remaining load and safety checks are explicitly identified so the Estate Office can make a defensible selection after completing the field survey and obtaining manufacturer/material data.